KEYSIGHT

TECHNOLOGIES

CloudLens Stack

Deployment Runbook for AWS

vContoller, KVO, vPB and sensors. End to end. One command.

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Keysight Technologies | Network Visibility Solutions

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1. Executive Summary

This runbook shows how to deploy the full CloudLens stack on AWS in one shot: vController (formerly CLMS), KVO (Keysight Vision One), the Virtual Packet Broker (vPB), and the OS sensors that connect tagged EC2 instances back to vController. Three paths are documented; pick the one that matches how your team works. All three produce the same AWS resources.

Audience

- Customer DevOps and platform engineers who own the AWS account.
- Keysight Sales Engineers running customer POCs or demos.
- Customer CTO / procurement / training teams who need a printable reference.

What gets deployed

Component	AWS resources	Required
vController (CLMS)	1 EC2 (t3.xlarge), 1 Elastic IP, security group, ENI	Yes
KVO (Keysight Vision One)	1 EC2 (c5.2xlarge), 1 Elastic IP, security group, ENI	Optional
Virtual Packet Broker (vPB)	1 EC2 (t3.xlarge), 1 Elastic IP, security group, ENI	Optional
Collector SVM	Auto-deployed by KVO Zone Tapping when a Cloud Config commits	Auto
Shared networking	VPC 10.99.0.0/16 with mgmt, data and tool subnets, IGW, route table	Yes

Time and cost

Phase	Wall-clock time
CloudFormation stack CREATE_COMPLETE	about 5 min
vController + KVO init (EULA reachable)	about 15 min
vPB reachable on port 9022	10 to 15 min
Sensors on first 10 instances (optional)	5 to 8 min
Total typical (stack + config + sensors)	30 to 45 min

Indicative AWS compute cost (us-east-1, June 2026 on-demand list): vController t3.xlarge and vPB t3.xlarge about USD 120/month each, KVO c5.2xlarge about USD 250/month, plus small EBS and Elastic IP fees. Stopping the instances stops compute charges. Marketplace software fees are billed separately by Keysight.

2. The Three Deployment Paths

All three paths use the same Marketplace AMIs, deploy the same EC2 instances, and chain vController first, then KVO and vPB. Pick the one that matches your operating model. You can switch between them later without losing state, because each path produces standard AWS resources in a CloudFormation stack or Terraform state.

Path	Best for	Tools needed	Time to first paste
A. Bash one-liner	First-time trial, customer demo, CloudShell users	Browser + AWS CLI (CloudShell has both)	10 seconds
B. Terraform stack module	IaC pipelines, repeatable customer envs, GitOps	Terraform 1.5+ and AWS CLI logged in	2 minutes
C. AWS Console (Launch Stack)	Click-through customers, governance-controlled accounts	Browser only	1 minute

Why three?

Customers do not all work the same way. A platform team wants Terraform. A network engineer trialling the kit wants a paste. A CTO walking through with procurement wants a button. Three paths, identical outcome.

3. Prerequisites

Confirm each item before kicking off the deployment. The script stops early with a clear message if any of the first three are missing, so you do not waste time mid-deploy.

☐ AWS account with rights to create a CloudFormation stack, VPC, EC2, and IAM resources (or AdministratorAccess for the SE-led path).

☐ AWS CLI v2 installed and logged in (SSO or access keys). In CloudShell this is preinstalled and pre-authenticated.

☐ Subscribed to the three Keysight Marketplace AMIs (Vision One, CloudLens Manager, Virtual Packet Broker). This is a one-time interactive step per account and cannot be automated.

☐ An EC2 key pair created in the target region (us-east-1 by default).

☐ vCPU quota headroom: c5.2xlarge (KVO, 8 vCPU) and t3.xlarge (vController and vPB, 4 vCPU each) in your region.

☐ Open egress on the workload subnets to reach vController on TCP/443 (only needed once sensors are deployed).

Region note

Default region is us-east-1 and the AMIs are region-locked. To deploy elsewhere, look up the equivalent Marketplace AMI IDs first: aws ec2 describe-images --owners aws-marketplace --filters Name=name,Values=*CloudLens*Manager* --region <region>. See docs/OPERATIONS.md section 9.

4. Path A: Bash One-Liner

The fastest way to a working stack. One paste in AWS CloudShell or any local terminal with the AWS CLI; the script drives CloudFormation and then the sensor playbook.

```
curl -sSL https://raw.githubusercontent.com/Keysight-Tech/cloudlens-ansible-aws/
main/deploy/deploy-stack.sh | bash
```

Prefer to inspect before running? Download then execute:

```
curl -sSL -o deploy-stack.sh \
  https://raw.githubusercontent.com/Keysight-Tech/cloudlens-ansible-aws/main/
deploy/deploy-stack.sh
less deploy-stack.sh
bash deploy-stack.sh --key-pair my-keypair
```

4.1 What the script does (phase by phase)

Each phase prints a banner with its name, so if anything fails you know exactly where you are.

1. Banner and environment detection: prints the version header and detects CloudShell vs local terminal so prompts adapt accordingly.
2. Pre-flight checks: confirms the AWS CLI is installed and you are logged in, and warns if the target region is missing quota for the c5 and t3 families.
3. Customer input: prompts for stack name (default cloudlens-stack), region (default us-east-1), EC2 key pair, and admin ingress CIDR. All defaults can be overridden by flags (see 4.2).
4. Marketplace check: reminds you to subscribe to the three Keysight AMIs if the account is not already subscribed. This one step is interactive on the Marketplace pages.
5. Stack deploy: submits deploy/cloudformation/stack.yaml. Waits for CREATE_COMPLETE and reads the Outputs (EIPs and private IPs).
6. Wait for init: polls the vController and KVO HTTPS endpoints until the UI is reachable (typically 15 minutes). Prints a one-line progress bar.
7. Manual project key step: the script pauses and tells you to open the vController UI, accept the EULA, create a project, and paste the project key back into the terminal. This is the one step that cannot be automated end to end today.
8. Sensor chain (optional): hands off to quickstart.sh, which deploys sensors to every EC2 instance tagged cloudlens=yes. Prints a final summary and writes cloudlens-deploy-summary.txt with all IPs and creds.

Output you can keep

Every run writes 'cloudlens-deploy-stack.log' (raw stdout/stderr) and 'cloudlens-deploy-summary.txt' (stack name, instance IDs, EIPs, default creds). Save both to your customer ticket or runbook archive.

4.2 Flags and overrides

Flag	Effect
--dry-run	Walk through every prompt and print the aws commands without touching AWS. Use this to preview the run.
--stack-name NAME	Override the default stack name (cloudlens-stack).
--region REGION	Override the default region (us-east-1). Pass a region where you have subscribed to the Marketplace AMIs.
--key-pair NAME	EC2 key pair for OS-level SSH access. Required.
--admin-cidr CIDR	Source CIDR allowed to reach SSH, vPB SSH (9022), HTTPS, and VXLAN. Narrow this from the 0.0.0.0/0 default.
--no-kvo / --no-vpb	Skip KVO or vPB. vController-only deploys complete fastest.
--no-sensors	Stop after the stack phase. Run quickstart.sh later yourself.
-h | --help	Print the help banner with all flags and a phase summary.

5. Path B: Terraform Stack Module

The stack module wraps the clms, kvo and vpb child modules behind one tfvars file. One 'terraform apply' provisions the VPC, subnets, security groups, ENIs, Elastic IPs and all three EC2 instances. The sensors are still deployed separately via Ansible because they touch customer instances, not AWS resources.

5.1 Workflow

9. Clone the repository and change into the stack module directory.
10. Copy terraform.tfvars.example to terraform.tfvars and fill in region, key_pair_name and admin_cidr. Toggle deploy_kvo / deploy_vpb if you only want vController.
11. Run terraform init to download the aws provider.
12. Run terraform plan to review the resources Terraform will create. Confirm against your account's policies.
13. Run terraform apply. About 5 to 10 minutes later, the stack is up. The outputs print the vController, KVO and vPB URLs and the default creds.
14. Open the vController UI, accept the EULA, create your project, copy the project key, and run quickstart.sh to deploy sensors.

```
git clone https://github.com/Keysight-Tech/cloudlens-ansible-aws.git
cd cloudlens-ansible-aws/deploy/terraform/stack
cp terraform.tfvars.example terraform.tfvars
$EDITOR terraform.tfvars   # set region, key_pair_name, admin_cidr
terraform init
terraform plan
terraform apply
```

5.2 What gets created

Resource	Count	Notes
VPC	1	10.99.0.0/16 (override via tfvars)
Subnets	3	mgmt 10.99.1.0/24, data 10.99.11.0/24, tool 10.99.12.0/24
Internet gateway + route table	1	Public egress for mgmt subnet
Elastic IPs	1 to 3	One per deployed component
Security groups	1 to 3	Scoped to the admin CIDR per role
EC2 instances	1 to 3	vController t3.xlarge, KVO c5.2xlarge, vPB t3.xlarge

Existing VPC?

The child modules (deploy/terraform/clms, kvo, vpb) can each target an existing VPC and subnet if you pass vpc_id and subnet_id instead of letting the stack create the shared VPC. Use this when the customer mandates bring-your-own-network.

6. Path C: AWS Console (Launch Stack)

If you prefer to deploy from the Console, the same CloudFormation template is exposed through the Launch Stack button on the README and the landing page at keysight-tech.github.io/cloudlens-ansible-aws. The button opens the CloudFormation quick-create screen with the template URL pre-loaded from this repository.

15. Click Launch Stack. Sign in to the AWS Console if prompted.
16. The quick-create form opens with the stack template already loaded from the repo's raw GitHub URL.
17. Fill the parameters: EC2 key pair, admin ingress CIDR, and whether to deploy KVO and vPB (both default to yes). Instance types are fixed and should be left at their defaults.
18. Tick the IAM capability acknowledgement and click Create Stack.
19. Wait about 5 minutes for CREATE_COMPLETE, then read the Outputs tab for the vController, KVO and vPB URLs and IPs.
20. Open the vController UI, accept the EULA, create a project, copy the key, and run quickstart.sh from CloudShell to deploy sensors.

Marketplace subscription first

The stack launches Marketplace AMIs. If the account has not subscribed to Keysight Vision One, CloudLens Manager, and CloudLens Virtual Packet Broker, the instances fail to launch. Subscribe once per account on the Marketplace listing pages before clicking Launch Stack.

No screenshots, by design

The AWS Console shifts every few months. Rather than ship stale screenshots, this guide describes the click path in words. The buttons and field names match the Console at June 2026.

7. Verification Checklist

Walk this list after every deploy. If any item fails, jump to section 8 for the matching troubleshooting entry.

☐ CloudFormation stack shows CREATE_COMPLETE and the Outputs tab lists the vController, KVO and vPB EIPs.

☐ vController UI reachable on https://<vcontroller-eip>/. The Keysight CloudLens login screen loads, not a connection error.

☐ Logged in to vController with admin / Cl0udLens@dm!n and completed the forced password change.

☐ Project created in vController under Settings > Projects, and the API key copied into a safe place.

☐ vPB reachable on SSH port 9022 after 10 to 15 minutes: ssh -p 9022 admin@<vpb-eip> (password ixia, then change it).

☐ KVO > Inventory > Devices shows vController and vPB as CONNECTED after adoption.

☐ Sensors visible in vController > Sensors within 8 minutes of running quickstart.sh, each showing Connected.

Printable

This page is intentionally a single checklist so it can be printed and ticked off during a customer handover or installation walkthrough.

8. Troubleshooting Reference

Organised by failure mode. The full reference lives at docs/OPERATIONS.md and docs/TROUBLESHOOTING.md in the repository.

Marketplace and instance types

Symptom	Likely cause	Fix
Stack creates but instances fail to launch	Not subscribed to the Marketplace AMIs	Subscribe once per account to Vision One, CloudLens Manager, and vPB
UnsupportedOperation on create	Wrong instance type for an AMI	KVO must be c5.2xlarge; vController and vPB must be t3.xlarge

IAM and quota

Symptom	Likely cause	Fix
AccessDenied on stack create	Principal cannot create IAM resources	Deploy with CAPABILITY_NAMED_IAM (CFN) or AdministratorAccess
InsufficientInstanceCapacity / quota error	Region quota too low for c5 or t3	Request a quota increase, or deploy in a region with headroom

Network

Symptom	Likely cause	Fix
vController UI not reachable after 20 minutes	Security group blocks 443 from your client IP	Narrow but include your IP in the admin ingress CIDR
vPB SSH times out on 9022	Security group missing TCP/9022, or KCOS still booting	Add the SG rule; wait 10 to 15 min after the instance is running
Sensor cannot reach vController	Egress blocked from instance subnet to vController on 443	Open egress on the workload subnet or peer the VPCs

Timing

Symptom	Likely cause	Fix
vController UI returns 502 / 503 just after deploy	Services still initialising in sequence	Wait the full 15 minutes; the script's poll loop handles this
KVO EULA blocks the API	EULA not accepted in a browser	Open https://<kvo-eip>/ and click Agree; the CLI cannot do it

9. Cleanup / Decommission

Two ways to undo a stack deploy. Pick the one that matches how you deployed it.

Terraform deploy

If you used Path B (Terraform), use terraform destroy. This removes exactly what Terraform created and leaves anything else in the account untouched.

```
cd deploy/terraform/stack
terraform destroy
```

Bash or Console deploy

If you used Path A or Path C, delete the CloudFormation stack. This removes the VPC, instances, Elastic IPs and security groups the stack created. Double-check the stack name before pressing enter.

```
aws cloudformation delete-stack --stack-name cloudlens-stack --region us-east-1
aws cloudformation wait stack-delete-complete --stack-name cloudlens-stack --
region us-east-1
```

Sensors on workload instances

Sensors are not removed by terraform destroy or delete-stack, because they live on the customer's workload instances (usually a different account or VPC). To remove them:

- Linux: docker stop cloudlens-agent && docker rm cloudlens-agent
- Windows: Stop-Service CloudLensAgent; sc.exe delete CloudLensAgent
- Or: untag the instances (remove cloudlens=yes) and rerun cleanup.yaml

Marketplace billing

Deleting the stack stops the Keysight Marketplace subscription billing for those instances at the next hourly tick. Compute billing stops immediately. Allow up to 24 hours for the Marketplace line to disappear from your AWS bill.

10. Appendix: File Paths and Quick Links

Repository file paths

File	Purpose
deploy/deploy-stack.sh	Path A entry point (Bash one-liner)
deploy/cloudformation/stack.yaml	Stack template (Paths A and C)
deploy/terraform/stack/	Path B entry point (Terraform module)
deploy/terraform/{clms,kvo,vpb}/	Per-component Terraform child modules
docs/CloudLens_Stack_Deployment_Runbook.pdf	This document (PDF)
docs/CloudLens_Stack_Deployment_Runbook.docx	This document (Word)
quickstart.sh	Sensor deployment chain (called by deploy-stack.sh)
demo/setup-aws-visibility-demo.sh	Full demo orchestrator (SE lab)

Command reference

Action	Command
Run full stack (recommended)	curl -sSL https://raw.githubusercontent.com/Keysight-Tech/cloudlens-ansible-aws/main/deploy/deploy-stack.sh | bash
Run full stack (dry-run)	bash deploy/deploy-stack.sh --dry-run
Run vController only	bash deploy/deploy-stack.sh --no-kvo --no-vpb
Terraform apply	cd deploy/terraform/stack && terraform apply
Terraform destroy	cd deploy/terraform/stack && terraform destroy
Delete the stack	aws cloudformation delete-stack --stack-name cloudlens-stack --region us-east-1

Useful links

Resource	Location
GitHub repository	https://github.com/Keysight-Tech/cloudlens-ansible-aws
Landing page	https://keysight-tech.github.io/cloudlens-ansible-aws/
Sensor runbook (companion document)	docs/CloudLens_Ansible_AWS_Customer_Runbook.pdf
Operations guide	docs/OPERATIONS.md
Troubleshooting reference	docs/TROUBLESHOOTING.md
Issues / support	https://github.com/Keysight-Tech/cloudlens-ansible-

	aws/issues

Support email subject (for the Keysight account team): 'CloudLens Stack AWS: <customer name>, <brief issue>'.